EE782: Assignment 1

By. Vishal Chandra Patel
23B0745

Summary

In this assignment, we successfully implemented end-to-end image captioning for remote sensing imagery using both CNN + LSTM and CNN + Transformer architectures. The workflow included:

1.Data preprocessing: Resizing images, tokenizing captions, building a word-level vocabulary, and generating dataset statistics.

2.Model building: Implementing CNN encoders (ResNet-18, MobileNet) and decoders (LSTM and Transformer), with appropriate projection of image features.

3.Training & debugging: Handling typical issues like padding mismatches, masking, shape conflicts, device placement, loss computation, and decoding errors.

4.Experiments: Backbone comparisons, rotation-invariant augmentation, and decoding strategies.

5.Evaluation & explainability: Quantitative metrics (BLEU-4, METEOR), qualitative analysis, Grad-CAM visualizations, attention maps, and mis-caption case studies.

The assignment emphasized LLM-aware development, where each coding error was documented, analyzed, and fixed with proper unit checks.

Conclusion

This assignment demonstrates that modular design, careful debugging, and thorough evaluation are essential for building robust image captioning systems in remote sensing. Key takeaways include:

1.CNN + LSTM and CNN + Transformer can both generate meaningful captions, but Transformer provides better context modeling and attention visualization.

2.Proper handling of padding, masks, shapes, and device placement is crucial to avoid runtime errors.

3.Feature projection, gradient management, and decoding strategy significantly affect model stability and output quality.

4.Explainability techniques like Grad-CAM and token occlusion help understand model decisions and identify failure cases.

5.Documenting LLM-assisted fixes enhances reproducibility and understanding of model behavior.

Overall, this project strengthened skills in deep learning for vision and NLP integration, debugging complex pipelines, and evaluating model performance beyond standard metrics, while providing a framework for future improvements in remote sensing captioning tasks.